MIL-C-19902C
30 September 1986
SUPERSEDING
MIL-C-19902B
23 July 1979

MILITARY SPECIFICATION

CAPS, VENT, FUEL STORAGE TANK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of fuel storage tank vent caps with screen covered vent openings. The vent cap fits on the vent pipe by means of a threaded inlet or slip-over inlet.

1.2 Classification. Vent caps covered by this specification shall be of the following types, sizes, and styles as specified (see 6.2.1 and 6.4).

TYPES

Type I - Vapors are directed downward through either two screened outlets or sectionalized outlets.

Type II - Vapor recovery vent cap. Vapors are directed upward through a screened circular outlet. Rain is prevented from entering the screened outlet by a rainhood with drain spouts. The cap is equipped with a 1/2-inch restricted opening for vapor recovery.

Type III - Vapors are directed upward through a screened circular outlet. Rain is prevented from entering the screened outlet by a rainhood with drain spouts.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5430

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Type IV - Vapor recovery vent cap. Vapors are directed upward through a screened circular outlet. Rain is prevented from entering the screened outlet by a rainhood with drain spouts. The cap is equipped with a spring loaded pressure vacuum poppet for vapor recovery.

SIZES

Size 1 - For 1-1/4 inch iron pipe size (ips) National Pipe Taper Thread (NPT) vent pipe (size 1 not applicable for Type IV cap).

Size 2 - For 1-1/2 inch ips NPT vent pipe (size 2 is not applicable for the Type IV cap).

Size 3 - For 2 inch ips NPT vent pipe (size 3 applicable to all types of caps specified herein).

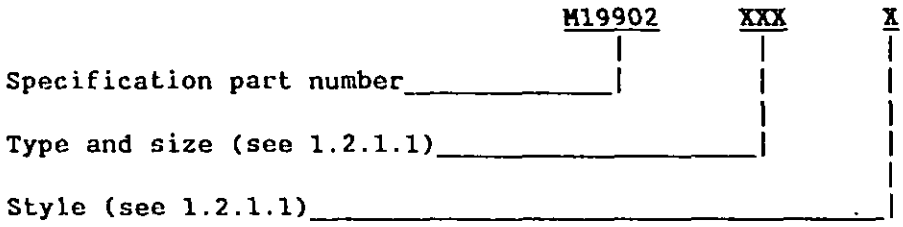
Size 4 - For 3 inch ips NPT vent pipe (size 4 not applicable for Type IV cap).

STYLES

Style A - Threaded inlet.

Style B - Slip-over inlet.

1.2.1 Specification part number. The specification part numbers for the caps covered by this specification are established for the types, sizes, and styles as follows:



1.2.1.1 Types, sizes, and styles. The type and size is identified by a three digit number. The first digit identifies the type and the last two digits identify the size. The style is identified by a single alpha character. Table 1 identifies the part number.

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TABLE I. Type, size, and style to code number.

Type and size	Style	Code Number
Type I		
Size 1	A	101A
	B	101B
Size 2	A	102A
	B	102B
Size 3	A	103A
	B	103A
Size 4	A	104A
	B	104B
Type II		
Size 1	A	201A
	B	201B
Size 2	A	202A
	B	202B
Size 3	A	203A
	B	203B
Size 4	A	204A
	B	204B
Type III		
Size 1	A	301A
	B	301B
Size 2	A	302A
	B	302B
Size 3	A	303A
	B	303B
Size 4	A	304A
	B	304B
Type IV		
Size 3 1/	A	403A
	B	404B

1/ The type IV is available only in the size 3.

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1.2.1.2 Interchangeability of part numbers. For binning purposes, the part numbers of the previous revision that are interchangeable with the part numbers of this revision are shown in Table II.

TABLE II. Interchangeability of part numbers.

New Part Number	Interchangeable with
M19902-101A	M19902-01A
M19902-102A	M19902-02A
M19902-103A	M19902-03A
M19902-201A	M19902-04A
M19902-201B	M19902-04B
M19902-202A	M19902-05A
M19902-202A	M19902-05B
M19902-203A	M19902-06A
M19902-203B	M19902-06B

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-V-3 - Valves, Fittings and Flanges (Except for Systems Indicated Herein) Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services Section 7, Pipe Threads, General Purpose.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A153 - Zinc-Coating (Hot Dip) on Iron and Steel Hardware.
B580 - Anodic Oxide Coatings on Aluminum.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

30 - Flammable and Combustible Liquids Code.

(Applications for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard commercial product. Each vent cap of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this

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specification, but which are a part of the manufacturer's standard commercial product, shall be included in the vent cap being furnished. Standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection and approval (see 4.2.1 and 6.3).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specifically specified.

3.3.1 Vent cap body. Unless otherwise specified (see 6.2.1), the vent cap body shall be fabricated from aluminum or cast iron. When aluminum is used, the vent cap body shall be zinc-coated in accordance with ASTM B580, type A, and when cast iron is used the vent cap body shall be zinc-coated in accordance with ASTM A153. The vent cap body shall be so treated as to be thoroughly coated with a smooth, bright adherent solid surface of zinc and shall be zinc-coated both inside and outside. The vent cap shall be coated before threading.

3.4 Interchangeability. All vent caps of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Construction.

3.5.1 Type I. The Type I vent cap shall be of construction similar to the illustrations shown in Figures 1 and 2. The pipe (inlet) end shall provide for attachment to the vent pipe by either internal NPT for Style A, or by a slip-over fit for Style B. The Style A vent shall be provided with flat surfaces or lugs at the base for tightening with a smooth jaw or spanner wrench. The Style B shall be provided with set screw(s) for tightening onto the vent pipe. The tightening set screw(s) on the Style B vent shall be of corrosion-resistant steel. The vent outlet(s) shall be of proportionate size to the inlet and fitted with a screen. The screen shall be 40 mesh size of brass or stainless steel. The screen shall be held securely in position.

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3.5.2 Type II. The type II vent cap shall be of construction similar to the illustration shown in Figure 3. The pipe (inlet) end shall provide for attachment to the vent pipe by either internal NPT for Style A, or by a slip-over fit for Style B. The Style A vent shall be provided with flat surfaces or lugs at the base for tightening with a smooth jaw or spanner wrench. The Style B shall be provided with set screw(s) for tightening onto the vent pipe. The tightening set screw(s) on the Style B vent shall be of corrosion-resistant steel. The vent outlet(s) shall be of proportionate size to the inlet, fitted with a screen cover. The screen cover shall be 40 mesh size of brass or stainless steel and held securely in position. The 1/2-inch restricted opening for the vapor recovery connection shall be provided with a plug or closure when the opening is not used in the installation. The rainhood shall be constructed so as to prevent entry of rain into the vent outlet. The vent shall be constructed so as to direct vapors upward in conformance with NFPA 30.

3.5.3 Type III. The type III vent cap shall be of construction similar to the illustration shown in Figure 4. The pipe (inlet) end shall provide for attachment to the vent pipe by either internal NPT for Style A, or by a slip-over fit for Style B. The Style A vent shall be provided with flat surfaces or lugs at the base for tightening with a smooth jaw or spanner wrench. The Style B shall be provided with set screw(s) for tightening onto the vent pipe. The tightening set screw(s) on the Style B vent shall be of corrosion-resistant steel. The vent outlet(s) shall be of proportionate size to the inlet, fitted with a screen cover. The screen cover shall be 40 mesh size of brass or stainless steel and held securely in position. The rainhood shall be constructed so as to prevent entry of rain into the vent outlet. The vent shall be constructed so as to direct vapors upward in conformance with NFPA 30.

3.5.4 Type IV. The type IV vent cap shall be of construction similar to the illustration shown in Figure 5. The pipe (inlet) end shall provide for attachment to the vent pipe by either internal NPT for Style A, or by a slip-over fit for Style B. The Style A vent shall be provided with flat surfaces or lugs at the base for tightening with a smooth jaw or spanner wrench. The Style B shall be provided with set screw(s) for tightening onto the vent pipe. The tightening set screw(s) on the Style B vent shall be of corrosion-resistant steel. The vent outlet(s) shall be of proportionate size to the inlet, fitted with a screen cover. The screen cover shall be 40 mesh size of brass or stainless steel and held securely in position. The pressure setting shall be either 8 ounces, 12 ounces, or 16 ounces as specified (see 6.2.1). The vacuum setting shall be 1/2 ounce. The rainhood shall be constructed so as to prevent entry of rain into the vent outlet. The vent shall be constructed so as to direct vapors upward in conformance with NFPA 30.

3.5.5 Adapter. When the 1-1/4 inch vent cap is required but is not available, and when specified (see 6.2.1), a threaded reducer may be substituted as part of the vent cap to adapt the 1-1/2 inch vent cap to the 1-1/4 inch vent pipe. Unless otherwise specified (see 6.2.1), the adapter shall be fabricated from corrosion-resistant metal or cast iron which is zinc coated to prevent corrosion.

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3.5.6 Threads. All pipe connection and screw threads shall be right hand and conform to FED-STD-H28.

3.6 Identification marking. Identification shall be permanently and legibly marked directly on the vent cap or on a corrosion-resisting metal plate securely attached to the vent cap at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name, and trade mark to be readily identifiable to the manufacturer.

3.7 Workmanship.

3.7.1 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.7.2 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one complete vent cap when a first article sample is required (see 3.2 and 6.2.1). This inspection shall include the examination of 4.4, and, when specified, the preproduction pack inspection of 4.5 (see 4.5 and 6.2.1). The first article may be either a first production item or a standard production item from the supplier's current inventory, provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4 and the packaging inspection 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete vent cap. All vent caps offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the Acceptable Quality Level shall be 4.0 percent defective. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. The first article, when furnished, and each sample selected in accordance with 4.3 shall be examined for compliance with the requirements in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute one defect.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-V-3 specified for fittings. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2.1), a preproduction pack shall be furnished for examination within the time frame required (see 6.2.1).

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5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-V-3, for fittings, with the level of preservation and packaging and the level of packing as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The types I and III vent caps are intended for venting of fuel storage tanks where vapor recovery is not required. The type I is intended for venting to the atmosphere and venting downward is permitted and the type III is intended for venting to the atmosphere and venting upward is required. Types II and IV are intended for venting of fuel storage tanks where vapor venting is required to be upward and vapors are to be recovered. Venting is released to the atmosphere or recovered by provisions incorporated in the design of the types II and IV for connecting to vapor recovery systems when vapor recovery during filling operations of fuel storage tanks are required. The type IV is available only in the size 3 (2-inch). Style A could also be for JP-5, whereas Style B would have to be used for the low flash fules like gasoline or JP-4. These fill caps should only be used on jet fuel tanks which have submerged fill lines to prevent the generation of static electricity.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and data of this specification.
b. Type, size, and style of vent cap required (see 1.2).
c. Type of material for vent cap body, if other than as specified (see 3.3.1).
d. Pressure setting for the type IV vent cap (see 3.5.4).
e. When an adapter is required (see 3.5.5).
f. When a first article sample and inspection is required. When a preproduction pack inspection is required (see 4.2.1 and 4.5).
g. Time frame required for preproduction pack inspection (see 4.5)
h. Level of preservation and level of packing (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross reference to classification of previous revisions. The following is a historical record of classification changes made in previous revisions of this specification.

MIL-B-19902C	MIL-B-19902B
Type I	Type I
Size 1	Size 1
Size 2	Size 2
Size 3	Size 3
Size 4	not designated
Style A	Style A
Style B	not designated
Type II	Type II
Size 1	Size 4
Size 2	Size 5
Size 3	Size 6
Size 4	not designated
Style A	Style A
Style B	Style B
Type III	not designated
Size 1	not designated
Size 2	not designated
Size 3	not designated
Size 4	not designated
Style A	not designated
Style B	not designated
Type IV	not designated
Size 3	not designated
Style A	not designated
Style B	not designated

6.5 Subject term (key word) listing.

Caps, vent, fuel storage tank
Vacuum pressure poppet
Vapor recovery
Vented

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Navy - YD

Review Activity:

DLA - CS

Preparing Activity:

Navy - YD

(Project 5430-0162)

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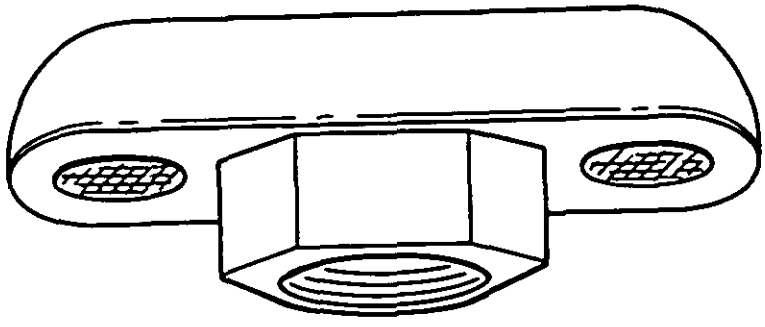


FIGURE 1. Type I vent cap, double outlet, downward venting.

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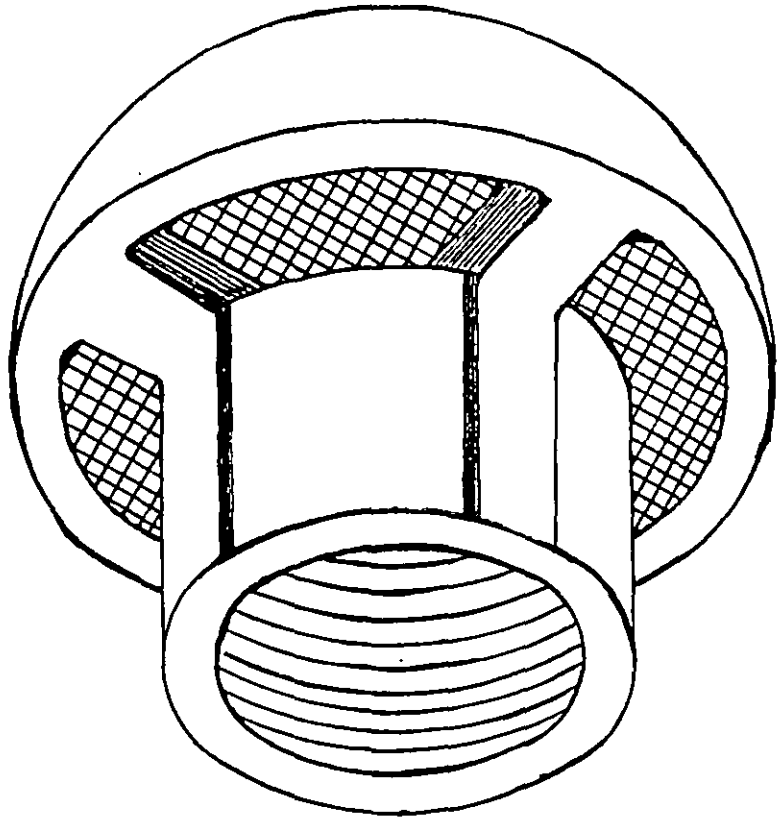


FIGURE 2. Type I vent cap, sectionalized outlet, downward venting.

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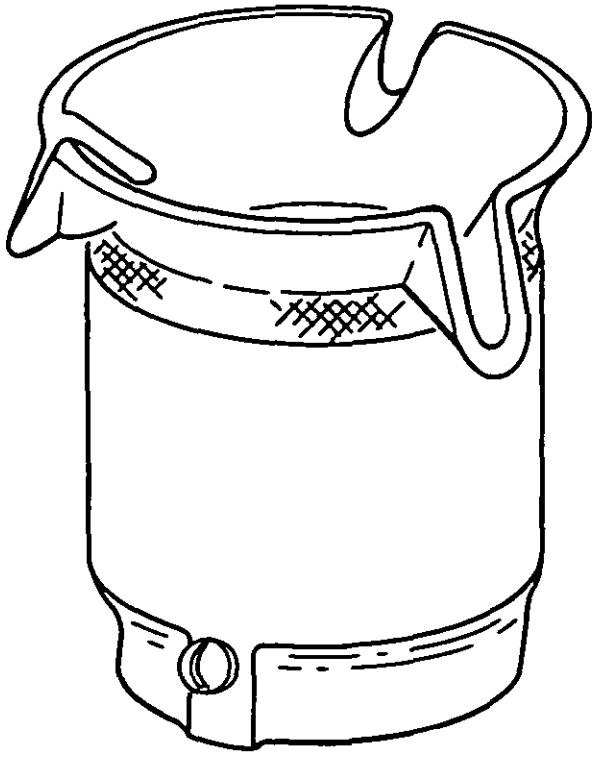


FIGURE 3. Type II vent cap, upward venting with 1/2-inch opening for vapor recovery.

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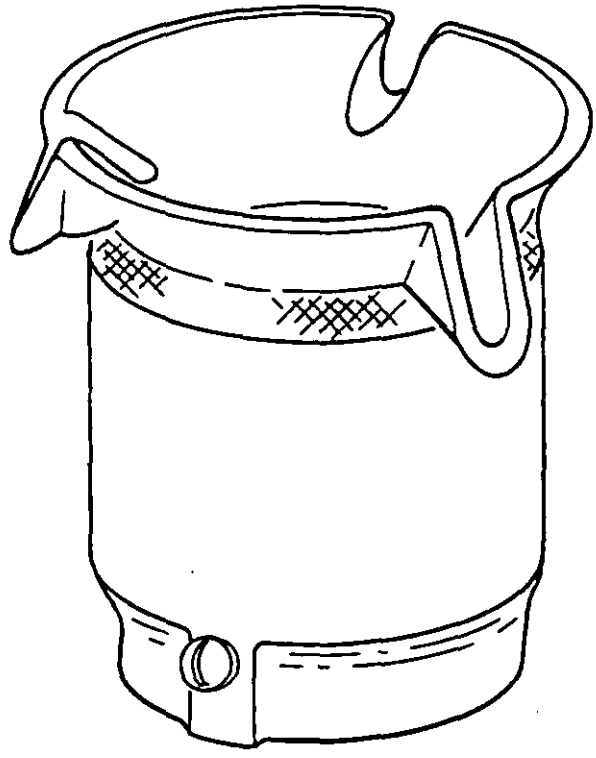


FIGURE 4. Type III vent cap, upward venting.

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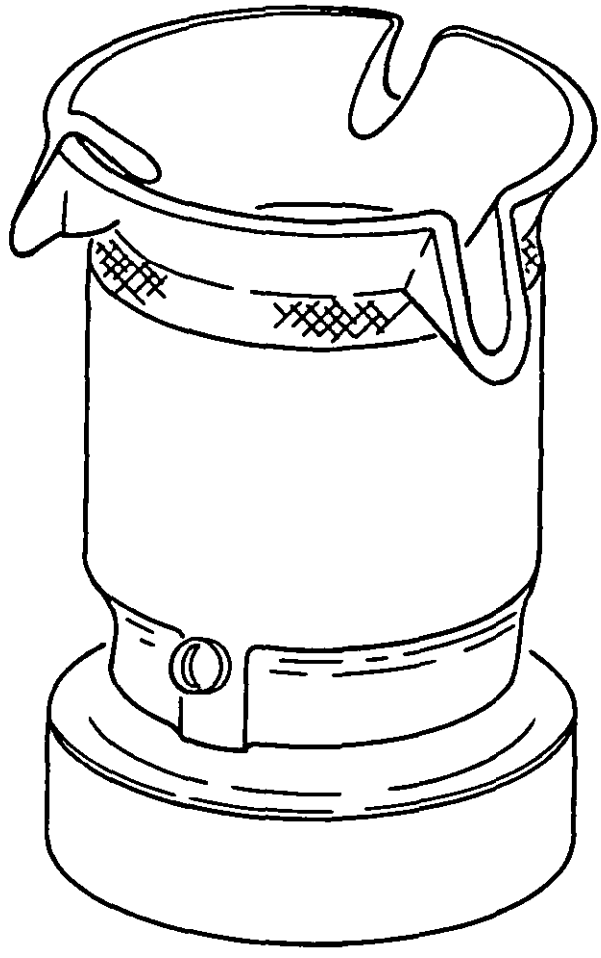


FIGURE 5. Type IV vent cap, upward venting with pressure vacuum poppet for vapor recovery.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-19902C	CAPS, VENT, FUEL STORAGE TANK

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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